On the Infeasibility of Constructing Lenstra–Pomerance Counterexamples to Agrawal's Conjecture via Twin-Smooth Subset Products

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Abstract

Agrawal's conjecture asserts that for a prime $r \nmid n$, the congruence $(X-1)^n \equiv X^n - 1 \pmod{n,\, X^r-1}$ forces n prime or $n^2 \equiv 1 \pmod r$; if true it yields a deterministic $\tilde{O}(\log^3 n)$ primality test. The only known route to a counterexample is the Lenstra–Pomerance construction: a squarefree n whose prime factors p satisfy $p-1 \mid n-1$ and $p+1 \mid n+1$ with auxiliary local conditions. We report on a sustained attempt to *construct* such an n of unrestricted size, as a two-stage pipeline: harvest a pool of primes $p = 2m+1$ with $(m, m+1)$ twin-smooth over partitioned factor bases, then solve a two-sided 0/1 subset-product in $(\mathbb{Z}/\Lambda_-)^\times \times (\mathbb{Z}/\Lambda_+)^\times$. The supply stage is solved: a partition-first harvester achieves measured supply/demand ratios of 13.8–209 against a yield model validated (conservatively) on the complete $B = 547$ twin-smooth corpus. The solve stage is blocked, by a *measured, two-part obstruction*: pools dense enough for the only scalable subset-product solver family (the Alford–Granville–Pomerance / Löh–Niebuhr / AGHS methods, which need identity-density $\rho \gtrsim 0.98$ by a measured generalized-birthday reach law $r_{\max} \sim \log N/(1-\rho)$) are unharvestable under the double condition, while harvestable pools have $\rho \approx 0.004$. We then show the obstruction is structural rather than parametric: a torus dimension-count theorem proves the double (twin-smooth) condition is the unique minimum-burden cell for *every* AKS modulus r; Popovych's two-section variant is strictly harder (burden ≥ 3 everywhere, 4 at $r = 5$); and a stabilizer-index theorem extends the same floor to the CM-elliptic analogue. The structural statements are unconditional modulo one named equidistribution-type hypothesis — Artin-type *generic sharpness*, the attainment of the proved support bounds — empirically supported (zero exceptions across 196 sampled large primes) and a theorem in the function-field mirror by Weil's RH for curves; the restriction to Frobenius-matching mechanisms is a separate, stated scope. This is not a disproof of the conjecture, nor a proof that no counterexample can be constructed; it is a quantified closure of the known constructive route, with the residual open directions stated precisely.

1 Introduction

For coprime integers n, r write $T(-1, n, r)$ for the congruence

$$(X-1)^n \equiv X^n - 1 \pmod{n,\, X^r-1}. \tag{1}$$

Conjecture 1.1 (Agrawal; Bhattacharjee–Pandey [2])**.** *If r is a prime not dividing n and $T(-1, n, r)$ holds, then n is prime or $n^2 \equiv 1 \pmod r$.*

If true, a single evaluation of (1) at one $r = O(\log n)$ would certify primality deterministically in $\tilde{O}(\log^3 n)$ time, against $\tilde{O}(\log^6 n)$ for the best proven algorithm [4]. The conjecture's status is

peculiar: Lenstra and Pomerance [3] gave a heuristic argument for $\gg x^{1-\varepsilon}$ counterexamples below x, later extended by Popovych [5] and Hegde–Devaraj [7], yet no counterexample is known, exhaustive search is clean to 10^{17} (Primaboinca, 2010–2020), and folklore places the least counterexample at hundreds of digits.

Searching is therefore hopeless, but *constructing* is not obviously so: the Lenstra–Pomerance (L–P) mechanism is deterministic — when its arithmetic side conditions hold, (1) follows with no further luck — and its heuristic density *improves* with size. This paper reports the outcome of carrying that program through, with every stage instrumented and measured. Our contributions:

(i) **Supply is solved.** A *partition-first* harvester for the required pool primes achieves measured supply/demand ratios of 13.8–209 (gate threshold 2.5), with a per-candidate yield model validated against the complete $B = 547$ twin-smooth corpus (predicted/actual 0.41–1.04, geometric mean 0.68: conservative). Along the way we quantify why the naive approach fails: mining any fixed twin-smooth corpus hits a measured 0.06 *invariant* (≈ 0.06 usable elements per demand bit, a factor 16–25 below feasibility), stable across corpora from 78 to 73,163 pool elements (§3).

(ii) **The solver stage is blocked by a measured density gap.** The group's 2-part is GF(2)-linear and solved at full scale, but carries only ∼16% of the constraint bits. For the odd part, generic solvers cap 2–3 orders of magnitude below the required dimension; the only scalable family (AGP/Löh–Niebuhr/AGHS subset-product methods [8, 9, 10]) requires pool identity-density $\rho \gtrsim 0.98$ by a measured generalized-birthday reach law, while pools harvestable under the double Carmichael + Lucas–Carmichael condition have $\rho \approx 0.004$, and pools with AGHS-grade density are measurably unharvestable (0–2 elements at every feasible scale). One can have a harvestable pool or a solvable one, not both (§4).

(iii) **The obstruction is structural, not parametric.** A torus dimension-count theorem shows that for *every* AKS modulus r the mechanism forces smoothness control on cyclotomic tori of total dimension ≥ 2, with the minimum 2 attained exactly by the split/nonsplit pair — the twin-smooth double condition — so no choice of r escapes (i); Popovych's variant is strictly harder (burden ≥ 3 everywhere, 4 at his own $r = 5$); and a stabilizer-index theorem extends the same floor to the CM-elliptic analogue of the test (§5).

(iv) **Reusable machinery and vocabulary.** A validated near-complete twin-smooth generator, the yield model, the density-reach diagnostic for Carmichael-type constructions, and the identification of the target objects with the *Williams numbers* of the generalized-Carmichael literature [17, 18, 19] (§7).

What is not claimed. This paper does not disprove Conjecture 1.1, which remains open and widely believed false. It does not prove that no counterexample can ever be constructed. Its negative results quantify the failure of the *known mechanism* (Frobenius-matching constructions in the L–P family) when driven by the *known solver families*; §6 states the boundary of the claim and the open directions precisely. The empirical statements are measurements, reproducible from the committed code and data (Appendix B); the structural theorems of §5 are proved, with one generic-sharpness ingredient whose status (heuristic, empirically supported, Artin-type) is flagged wherever used.

2 The construction and its reduction

Theorem 2.1 (Lenstra–Pomerance [3]; the $k \equiv 3 \bmod 4$ case in Váňa [6])**.** *Let $n = p_1 \cdots p_k$ with $p_1, \ldots, p_k$ distinct primes such that* (a) *k is odd;* (b) *$p_i \equiv 3 \pmod{80}$;* (c) *$p_i - 1 \mid n - 1$ for all i (n is Carmichael);* (d) *$p_i + 1 \mid n + 1$ for all i (n is Lucas–Carmichael). Then $T(-1, n, 5)$ holds and $n^2 \not\equiv 1 \pmod 5$; such an n with $k \geq 3$ is a counterexample to Conjecture 1.1.*

The mechanism behind Theorem 2.1 is *Frobenius matching*: modulo each $p \mid n$, the ring $\mathbb{F}_p[X]/(X^5 - 1)$ splits as $\mathbb{F}_p \times \mathbb{F}_{p^4}$, the Frobenius gives $(\zeta - 1)^{p^j} = \zeta^{p^j} - 1$ exactly, and conditions (b)–(d) force $n \equiv p \pmod{\operatorname{lcm}(5, \rho_5(p))}$, where $\rho_5(p)$, the multiplicative order of $X - 1$ in $(\mathbb{F}_p[X]/\Phi_5)^\times$, divides $10(p^2 - 1)$ [6]. Composite n built this way *impersonate the Frobenius* of each prime divisor on both one-dimensional algebraic tori over $\mathbb{F}_p$: the split torus ($\#\mathbb{G}_m(\mathbb{F}_p) = p - 1$, condition (c)) and the nonsplit norm-one torus ($\#T_2(\mathbb{F}_p) = p + 1$, condition (d)). In the vocabulary of the generalized-Carmichael literature, candidates satisfying (c)+(d) are exactly the *Williams numbers* [17, 18]: no example is known, none exists below 2^{64} [19] (nor below the Carmichael tables to 10^{18} [25]), and nonemptiness of the class is open — facts consistent with, and explained by, the measurements below.

Two elementary lemmas shape everything downstream.

Lemma 2.2 (side-coprimality)**.** *If n satisfies (c) and (d) then $\gcd(p_i - 1, p_j + 1) \mid 2$ for all i, j; the odd primes dividing $\prod_i (p_i - 1)$ and those dividing $\prod_i (p_i + 1)$ form disjoint sets.*

Proof. An odd prime q dividing $p_i - 1$ and $p_j + 1$ divides both $n - 1$ and $n + 1$, hence 2. □

Lemma 2.3 (twin-smooth dictionary)**.** *Let $p = 2m + 1$. Conditions (b) and the smoothness structure of $p \mp 1$ hold iff $m \equiv 1 \pmod{40}$, $(m, m + 1)$ is a pair of consecutive smooth numbers (a* twin-smooth *pair) with the odd part of m supported on one side Q_- and the odd part of $m + 1$ on the other side Q_+, and p is prime.*

Twin-smooth pairs with $2m+1$ prime are precisely the parameters sought by the isogeny-based schemes B-SIDH and SQIsign, so their generation is an industrial literature [12, 13, 15, 16] whose datasets we use throughout. Fixing finite disjoint odd prime sets Q_-, Q_+ (with $5 \notin Q_- \cup Q_+$), the pool is

$$\mathcal{P} = \left\{ p = 2m + 1 \text{ prime} : m \equiv 1 \pmod{40},\ m \text{ odd-}Q_-\text{-smooth},\ \tfrac{m+1}{2} \text{ odd-}Q_+\text{-smooth} \right\},$$

and with $\Lambda_- = 2 \operatorname{lcm}\{p - 1\}$, $\Lambda_+ = 4 \operatorname{lcm}\{p + 1\}$ over $\mathcal{P}$:

Proposition 2.4 (reduction to a two-sided subset product)**.** *If $S \subseteq \mathcal{P}$ with $|S|$ odd, $|S| \geq 3$, and $n = \prod_{p \in S} p$ satisfies*

$$n \equiv 1 \pmod{\Lambda_-}, \qquad n \equiv -1 \pmod{\Lambda_+}, \tag{2}$$

then n is a counterexample to Conjecture 1.1. The targets in (2) are consistent because $\gcd(\Lambda_-, \Lambda_+) = 2$ (Lemma 2.2).

Taking discrete logarithms in each cyclic component (Pohlig–Hellman is cheap: all component orders are smooth by design), (2) is a 0/1 subset-sum over $G = (\mathbb{Z}/\Lambda_-)^\times \times (\mathbb{Z}/\Lambda_+)^\times \cong \bigoplus_i \mathbb{Z}/m_i$, plus one $\mathbb{Z}/2$ parity coordinate. Pigeonhole solvability needs $|\mathcal{P}| \gtrsim \log_2 |G|$ with slack for the solver. The pipeline is therefore **harvest** ($\mathcal{P}$, with $|\mathcal{P}|/\log_2 |G|$ comfortably > 1) then **solve** (find S). The next two sections treat the stages in turn; all quantitative claims are measurements, traceable per Appendix B.

3 Supply is solved

3.1 Mining fixed corpora fails: the 0.06 invariant

The obvious approach is to mine existing twin-smooth datasets: filter by Lemma 2.3, then choose the side partition $Q_- \sqcup Q_+$ that retains the most elements (an element survives iff its m-support and $(m+1)$-support land on opposite sides). Measured across every scale available to us — including the *complete* $B = 547$ corpus of 82,026,426 pairs (courtesy of M. Naehrig), which yields 73,163 pool candidates after the congruence and primality filters — the best partition found by element-greedy search and by direct partition-space local search keeps

0.060–0.083 elements per demand bit

(Table 1), where demand is $\log_2$ of the group order the survivors generate. Feasibility needs ratio $\gtrsim$ 1–3: the shortfall is a factor 16–25, stable across two orders of magnitude in B-range and three in pool size, and insensitive to corpus completeness and to the optimizer used. Mechanism: a partition-agnostic twin pays ≈ 2 fresh primes $\times$ 7–7.5 bits of new demand per kept element. Dataset mining is dead as the primary route.

corpus	source	pool	best survivors	surv./demand-bit
$B = 113$, $X \le 10^8$	sieve	59	8	0.061
$B = 113$, $X \le 10^{10}$	sieve	73	8	0.061
$B = 113$, Størmer closure	CHM	78	8	0.061
$B = 257$, $X \le 10^9$	sieve	497	21	0.061
$B = 547$, $X \le 10^9$	sieve	2,715	51	0.068
$B = 200$, near-complete	CHM (ours)	562	17 / 16	0.060 / 0.061
$B = 547$, complete corpus	[13]	73,163	82	0.083

Table 1: The mining invariant. $B = 200$ lists element-greedy / partition-space search; the $B = 547$ row is the full published corpus (random-partition expectation there: 25.6 survivors).

As calibration infrastructure we implemented the Conrey–Holmström–McLaughlin closure [14, 13] in C; it reproduces the exact complete twin-smooth sets to 99.2–99.7% with zero false positives and exactly matching largest elements ($B = 100$: 13,333 of 13,374; $B = 200$: 346,110 against the true 348,840 — the originally published count 346,192 was itself 0.8% short), anchoring the supply curve at $N(B) \sim B^{\alpha}$ with fitted $\alpha \approx 4.7$–5.4.

3.2 Partition-first harvesting: gates G1 and G2 passed

The invariant is a global combinatorial tax, and it is bypassed by fixing the partition *first*: choose Q_-, Q_+, enumerate candidates m as products of j primes of Q_- (no smoothness luck needed on that side), and test only $(m+1)/2$ for Q_+-smoothness and $2m+1$ for primality. Every hit is partition-compatible by construction; the cost moves into per-candidate yield, a local Dickman-type probability. Measured ratios of projected pool to demand bits, at 8–20M samples per cell (Appendix A, Table 5): 13.8 at the frozen production spec (Q_+ = odd primes ≤ 1259 except 5, $Q_- \subset (1259, 12590]$, $j = 4$; pool $\approx 2.5 \times 10^5$ of mean 50-bit elements against $\approx 18{,}000$ demand bits) up to 209 at the widest measured cell — a factor 200–700 above the mining invariant, and 5.5–84$\times$ the go/no-go gate of 2.5. The underlying yield model was validated out-of-sample against the complete $B = 547$ corpus on four asymmetric splits: predicted/actual $0.71, 0.70, 0.41, 1.04$ (geometric mean 0.68) — *conservative, never optimistic.* A solver-aware co-design variant (both

factor bases restricted to primes q with $(q-1)$'s odd part 100-smooth) keeps ratio 7.7 while shrinking the solver group's odd-prime support from {336 primes ≤ 6269} to {24 primes ≤ 97}.

Supply, in short, is abundant and over-provisioned. The production harvest was intentionally never launched, because the binding constraint turned out to be downstream.

4 The solver obstruction

4.1 The easy part and the generic frontier

G decomposes into cyclic prime-power components. Its 2-part is GF(2)-linear (including the parity coordinate) and solves by exact Gaussian elimination at full scale: 4,586 components in 12 seconds at the frozen spec. But the 2-part carries only ∼16% of the demand bits; the remaining ∼84% (≈ 15,000 bits at the frozen spec) lies in odd-prime components with primes up to 6269 (up to 97 after co-design). On that odd part, generic methods hit a hard frontier, measured on faithful synthetic instances with planted solutions: simulated annealing inside the GF(2) solution family stalls at ≥19 odd components; exact meet-in-the-middle solves $N = 24$, $D = 11$ instantly and $N = 40$, $D = 15$ in 11 seconds but is exponential in N, capping near 40 pool elements and ∼15 odd components — 2–3 orders of magnitude below the thousands of components and ∼24,000 odd bits required.

4.2 The scalable family and what it needs

Subset products of exactly this shape are solved at astonishing scale in the Carmichael-construction literature: Löh–Niebuhr [9] and especially Alford–Grantham–Hayman–Shallue [10], who built a Carmichael number with 10,333,229,505 prime factors via a subexponential ω-guided subset-product reduction over $(\mathbb{Z}/\Lambda)^\times$ — including the full odd part, with no small-odd-part requirement. The method's engine is structural: it needs pool elements that are the identity on *many* CRT components — the "$p-1 \mid \Lambda$" divisor structure, identity-density $\rho \approx 0.3$–0.5 in AGHS pools.

Our harvestable pools do not have this structure. A partition-first pool prime $p = 2m+1$ with m a product of $j \approx 4$ medium primes is the identity on only $\sim j$ of the components: $\rho \approx 0.003$–0.008, two orders of magnitude below AGHS.

4.3 The two-part density gap

Both halves of the remaining question were measured.

High density is unharvestable (N1). The divisor-paradigm harvest that would supply AGHS-grade density — fix highly composite Λ_-, Λ_+ and demand $p-1 \mid \Lambda_-$, $p+1 \mid \Lambda_+$ — collapses under the double condition: across all feasible scales the pool is 0–2 primes (Table 2). The root cause is Lemma 2.2 made quantitative: the double condition is the conjunction of two rare events on *disjoint* fixed moduli, where AGHS needs only the single one-sided event $p-1 \mid \Lambda$.

Low density is unsolvable (A10). To close the remaining escape — perhaps the scalable solvers tolerate low density — we implemented a correct Wagner-4 generalized-birthday solver [11] and measured its *reach*: the largest component count $r_{\max}$ it solves, as a function of identity-density ρ and pool size N (Table 3). The measured law is

$$(1-\rho)\, r_{\max} \approx \text{const}, \qquad r_{\max} \sim \log N :$$

at the harvestable density $\rho \approx 0.004$ the solver reaches ∼20 components against the thousands required; reaching 1000 components needs $\rho \gtrsim 0.98$ (optimistic higher-k Wagner variants lower

T_- bound	T_+ range	$\lvert T_-\rvert + \lvert T_+\rvert$	pool	identity-density
90	97–220	45	1	0.178
120	127–300	60	1	0.067
150	160–360	68	2	0.074 (max 0.088)
60	67–160	34	0	—

Table 2: N1: the divisor-paradigm (AGHS-density) harvest under the double condition. Small bases give empty pools; larger bases collapse the density and the pool stays ∼empty.

this only to ≈0.85–0.95); and pool growth helps only logarithmically (5× the pool bought +4 components) while the group grows with the harvest, so the ratio never improves.

ρ	N	$r_{\max}$	$(1-\rho)\, r_{\max}$
0.004 (harvestable)	20,000	20	19.9
0.004 (harvestable)	100,000	24	23.9
0.5	20,000	28	14.0
0.9	20,000	128	12.8

Table 3: A10: measured Wagner-4 reach vs. identity-density.

Conclusion of the measurement campaign. Under the double Carmichael + Lucas–Carmichael condition one can have a harvestable pool *or* AGHS-grade identity density, **not both**: high-density pools are unharvestable (N1) and low-density pools are unsolvable by the known scalable family (A10). That two-part, quantified gap is the core negative result of this paper.

5 The obstruction is structural

The measurements of §4 concern $r = 5$ and the specific section $X - 1$. This section shows none of those choices can be varied away. The results are proved in the companion arithmetic-geometry document of this repository (`doc/arith-geom.tex`, v1.0); we state them with proofs sketched. One ingredient is conditional and we name it once, here: *generic sharpness* — the proved support bounds on the controlling orders are *attained* — an Artin-type equidistribution hypothesis, empirically supported (zero exceptions across 196 sampled large primes in three independent verification campaigns) and a *theorem* in the function-field mirror by Weil's RH for curves. The restriction to Frobenius-matching mechanisms (§6, item 2) is a stated scope, not a hypothesis; the two should not be conflated.

5.1 The stabilizer-index theorem

Modulo $p \mid n$, the mechanism's cost is governed by the multiplicative order of the test section in the component algebra, and that order collapses exactly when a Galois or CM symmetry fixes the section up to a root-of-unity cofactor. For $X - 1$, whose zero $\{1\}$ is Galois-stable, inversion gives the exact identity $(X-1)^{p^{d/2}-1} = -X^{-1}$ for even $d = \mathrm{ord}_r(p)$ (Váňa's $\rho_5(p) \mid 10(p^2-1)$ [6] is the case $d = 4$). For $X + 2$, whose zero -2 has trivial stabilizer, no collapse exists at any depth: $(\zeta^\gamma + 2)/(\zeta + 2)$ is not even a unit of $\mathbb{Z}[\zeta_r]$ for $\gamma \neq 1$ (disjoint ideal supports, $N(\zeta - 1) = r$ versus $N(\zeta + 2) = (2^r + 1)/3$). For the x-coordinate on a CM elliptic curve, the CM automorphisms give exact collapses such as $x([i]S) = -x(S)$ on $y^2 = x^3 - x$. All cases are one statement:

Theorem 5.1 (stabilizer-index floor; *proved*, sharpness generic)**.** *For a one-section Frobenius test on a one-dimensional group with scalar Frobenius action λ in a cyclic group and unit-stabilizer U (the subgroup acting on the section by root-of-unity cofactors), the smoothness burden a Frobenius-matching forgery must pay at each pool prime is the index*

$$k' \;=\; [\langle\lambda\rangle : \langle\lambda\rangle \cap U],$$

realized as simultaneous smoothness of the cyclotomic torus orders $\Phi_e(p)$ for $e \mid k'$ — total torus dimension k'. Every credible exception clause equals U, because U-cells are reachable by one-sided AGP-type constructions (§5.5); a valid counterexample therefore forces $\lambda \notin U$, hence ***burden*** *$k' \geq 2$.*

Proof sketch. With $e = k'$ minimal such that $\lambda^e \in U$, the cofactor identity $f^{p^{k'}} = \zeta f$ (ζ a root of unity of small order) bounds the order's support by the levels $e \mid k'$ — unconditionally, by the exact identities above. For $e < k'$ the cofactor $f \circ [\lambda^e]/f$ is a non-torsion unit (cyclotomic, or elliptic with disjoint divisor support), so no identity collapses further; generic sharpness then places the full levels $e \mid k'$ in the order. Validity excludes $\lambda \in U$ since U-cells pass only inside the exception clause. □

section	acting group	U	min. valid burden	realizability
$X - 1$ on μ_r	$(\mathbb{Z}/r)^\times$	$\{\pm 1\}$	2	$4 \mid r - 1$
$X + 2$ on μ_r	$(\mathbb{Z}/r)^\times$	$\{1\}$	3 (4 at r=5)	$3 \mid r - 1$ for 3
x on $E_{j=1728}$	eigenline scalars	μ_4	2	$8 \mid r - 1$
x on $E_{j=0}$	eigenline scalars	μ_6-image	2	$12 \mid r - 1$

Table 4: Instances of Theorem 5.1. Measured valid-cell floors on $y^2 = x^3 - x$: 3 at $r = 13$ and 2 at $r = 17$, matching the index formula.

The remainder of this section instantiates the theorem; each corollary was verified by its own deterministic script (Appendix B).

5.2 Every AKS modulus forces the double condition

Corollary 5.2 (torus dimension count)**.** *Take the section $X - 1$: $U = \{\pm 1\}$, so the burden at $p \mid n$ is $d'_p = d_p/2$ (d_p even) or d_p (d_p odd), where $d_p = \mathrm{ord}_r(p)$, and validity ($\mathrm{ord}_r(n) > 2$, which forces $d_p \geq 3$) gives burden ≥ 2 for* every *odd prime modulus r. The minimum 2 is attained exactly at $d_p = 4$, levels $\{1, 2\}$: $p - 1$ and $p + 1$ smooth — the double condition — realizable iff $4 \mid r - 1$. For $r \equiv 3 \pmod 4$ the burden is ≥ 3, i.e. smooth values of $\Phi_e(p) \sim p^{\varphi(e)}$ with $\varphi(e) \geq 2$ — the quantities whose non-smoothness underpins torus-based cryptography [24].*

The resulting burden formula, $\min_{d \mid r-1,\, d \geq 3}(d/2 \text{ if even else } d)$, reproduces an independent empirical scan over eleven moduli $r \leq 41$ exactly (value and minimizing d), and the collapse identity, exact orders and sharpness behaviour were verified on 67 sample primes — sampled and implemented independently of that scan (different ranges and residue-class selection; exact orders by polynomial arithmetic). **"Choose a friendlier r" is closed structurally**: every $r \equiv 1 \pmod 4$ reproduces the $r = 5$ economics identically, and every other r is strictly worse.

5.3 Popovych's variant is strictly harder

Corollary 5.3 (Popovych burden)**.** *Take the pair* $\{X-1, X+2\}$ *[5]: the second section has* $U=\{1\}$*, so the pair pays full depth* d_p *at every parity, and*

$$\mathrm{burden}_{\mathrm{Pop}}(r) = \min\{d : d \mid r-1,\ d \geq 3\} \ \geq\ 3 \quad \textit{for every } r, \qquad = 4 \ \textit{at } r=5,$$

i.e. at Popovych's own modulus the pool primes would need $p-1$*,* $p+1$ and p^2+1 *simultaneously smooth.*

At the frozen spec the extra ≈102-bit quadratic level is a pure Dickman tax $\rho(102/\log_2 B)$: 3.9×10^{-11} over the solver base $B=1259$, still 1.8×10^{-7} over the enumerated base $B'=12590$ — emptying the measured 2.5×10^5 pool to below one element; the solver-side group only grows. To our knowledge this is the first quantified evidence in either direction on Popovych's conjecture beyond exhaustive search; a self-contained treatment is the companion note `doc/popovych-note.tex`. The constant 2 is immaterial: for any integer $c \notin \{0, \pm1\}$ the same trivial-stabilizer argument gives the identical jump, while $c=\pm1$ restores the collapse — the zero $-c$ must avoid the rational torsion $\{\pm1\}$ of $\mathbb{G}_m$, and the innocent-looking $c=+1$ fails doubly ($\zeta+1$ is multiplicatively dependent on the $\zeta-1$ orbit, so $\{X-1, X+1\}$ adds no burden at all); $c=2$ is simply the smallest valid choice (companion note, final remark).

5.4 The CM-elliptic axis is closed

A canonical elliptic analogue of the substitution $X \mapsto X^n$ requires CM: a canonical commuting family of Frobenius lifts would make each ψ_p a degree-p endomorphism, and an endomorphism of non-square degree already forces $\mathrm{End}(E) \supsetneq \mathbb{Z}$; conversely for CM curves the canonical family exists [23]. There Frobenius acts on an r-eigenline by a scalar λ and Theorem 5.1 applies with U the image of the CM units:

Corollary 5.4 (elliptic closure)**.** *On the CM curves the apparent burden-*1 *cells (*λ *equal to a CM unit, e.g.* $\lambda \equiv \pm i$*, where the section order collapses to the split torus alone) lie exactly inside the exception clause — the elliptic mirror of the rigid cells of §5.5 — and the valid floor returns to* 2*, attained on levels* $\{1,2\}$*:* $p-1$ and $p+1$ smooth, the same twin-smooth demand*, with unit-twisted mixed targets reproducing the sign split. The N1/A10 obstruction of §4 therefore transfers verbatim.*

Measured on $y^2=x^3-x$: valid-cell floors 3 at $r=13$ and 2 at $r=17$, matching the index formula, with the elliptic burden ≥ the torus burden at all eleven moduli tested (and the test degenerate at $r=5$, where every cell is exceptional). Supersingular cells have $\varphi^2=[-p]$ and reduce to the torus analysis. The elliptic axis offers no single-rare-event escape.

5.5 The sign split, not the depth, is the wall

The generalized-Carmichael record explains *which* cells the known constructive machinery can reach, and confirms the theorem's shape empirically. Carmichael numbers occupy the cell (level $\{1\}$, Frobenius power 0) and are infinite [8]; Lucas–Carmichael numbers occupy $(\{2\}, 1)$ and are infinite [21] (via the elliptic Korselt criteria of [22]); the L–P targets occupy $(\{1,2\}, (0,1))$ — Williams numbers — empty to 2^{64} [19]; yet Howe's *rigid order-*2 *Carmichael number* 443,372,888,629,441 $= 17\cdot31\cdot41\cdot43\cdot89\cdot97\cdot167\cdot331$ [20] occupies the *same level set* with all-even powers ($p^2-1 \mid n-1$; re-verified by us) and exists at 4.4×10^{14}. Depth-2 impersonation is constructible; what is not is the *sign split*: the Williams powers $(0,1)$ force target $+1$ on the $p-1$ side and -1 on the $p+1$ side, which conflict on any shared prime power and so force the coprime side split of Lemma 2.2 —

exactly the structure whose harvest collapse is Table 2 and whose solver gap is Table 3. Rigid cells, by contrast, are one-sided AGP-type problems (single target 1, no coprimality demands), which is why the AGP machine reaches them — and why the exception clause of any credible conjecture in this family must contain the stabilizer cells, as Theorem 5.1 asserts. Validity ($n \not\equiv \pm 1 \bmod r$) pins the Agrawal construction to the sign-split cell: a rigid-cell n can pass (1) only with $n \equiv 1 \pmod r$, inside the exception clause — and indeed Howe's $n \equiv 1 \pmod 5$ does not even pass $T(-1, n, 5)$ (verified: check H4 of `code/ag3_hierarchy_check.py`), rigidity controlling $p^2 - 1$ but not the residual μ_{2r}-torsion that the mod-80 conditions exist to fix.

6 Scope, caveats, and what remains open

This section is the boundary of the result, not throat-clearing.

1. **Not a disproof.** Conjecture 1.1 remains open and believed false; nothing here bears on its truth. Likewise §5.3 is evidence about the known *mechanism* against Popovych's variant, not evidence that the variant is true.
2. **Mechanism scope.** All structural results quantify Frobenius-matching constructions ($n \equiv p^j \bmod \operatorname{lcm}(r, \rho_r(p))$ per prime). Every published counterexample approach [3, 5, 7] is an instance; "accidental" solutions of the component equations outside this family are not excluded, and no way to construct them is known.
3. **Generic sharpness.** The dimension-count and stabilizer-index theorems bound the support of the control orders unconditionally (exact collapse identities) but their *attainment* is an Artin-type genericity statement: a harvest could in principle fish in the thin set of primes whose section order collapses further (heuristic density $\sim 1/\ell$ per avoided large prime; empirically 0 exceptions across 196 sampled large primes in the three verification campaigns). That set is far thinner than the twin-smooth pool itself, so the escape is uneconomical, but it is not unconditionally excluded.
4. **Finite-scale caveats on N1/A10.** The density-gap measurements used squarefree divisor harvests and uniform small-prime component models at finite scale; prime-power moduli and enormous scales are untested, though the density trend and the two-rare-events argument point the same way, and the A10 reach law's logarithmic pool-dependence makes "outgrow it" quantitatively hopeless.
5. **What remains genuinely open.** (i) A fundamentally new solver paradigm for 0/1 subset-products in $\sim$24,000-bit smooth-exponent groups with dense vectors and only linear pool/dimension ratio — posed precisely, with a regression harness, in the repository's option-(b) brief. The results of §5 *sharpen rather than kill* this direction: a breakthrough there would unblock the torus route only, the Popovych and elliptic variants remaining strictly harder. (ii) The admissible-local-cells question (which residue cells mod 2^k and mod r support the mechanism — the boost polynomials $2x^n - 1$ are excluded by a mod-16 obstruction, but other side-separating families are not ruled out). (iii) The existence of Williams numbers at all, open independently of this program [19]. (iv) Unconditional versions of the sharpness statements — which, we note, are theorems in the function-field mirror by Weil's RH for curves, locating the residual difficulty over $\mathbb{Z}$ in the unconditional-equidistribution gap.

7 Reusable contributions and conclusion

Beyond the negative result, the campaign leaves usable artifacts: a validated C implementation of the CHM closure reproducing complete twin-smooth sets to 99.2–99.7% with exact extremes; a calibrated, conservative yield model for partition-first smooth-pair harvesting; an exact GF(2) solver for the 2-part of Carmichael-type subset products; the Wagner density-reach methodology as a *diagnostic* — given any proposed Carmichael-type construction, measure its pool's identity-density and read off the reach — and the impersonation-cell vocabulary tying this problem to the Williams/Howe literature, with the sign-split criterion predicting which cells the AGP machine can reach. All code and data are committed alongside this paper (Appendix B).

Two constructive reframings deserve emphasis. First, the stabilizer index of Theorem 5.1 is a *design metric* for AKS-style tests: for any proposed section (a polynomial section of μ_r, a CM x-coordinate) and any modulus, it computes the smoothness burden a Frobenius-matching forgery must pay — in effect a security parameter, maximized by choosing sections with small unit-stabilizer and moduli whose $(r-1)$-divisor structure keeps the index high (Table 4); Popovych's pair, read this way, is a deliberate stabilizer reduction from $\{\pm 1\}$ to $\{1\}$. Second, over $\mathbb{F}_q(T)$ (the Carlitz-module mirror) the sharpness hypothesis is a theorem by Weil's RH for curves, so the statement *"no Frobenius-matching forgery beats the stabilizer-index floor"* holds unconditionally there — a positive theorem that isolates the residual difficulty of the problem over $\mathbb{Z}$ in the unconditional-equidistribution gap.

The conclusion in one paragraph: the Lenstra–Pomerance route to an Agrawal counterexample is a two-stage pipeline whose supply stage we solved and whose solver stage is blocked by a measured identity-density gap — high-density pools are unharvestable under the double condition, low-density pools are unsolvable by every known scalable family — and the blockage is structural: the double condition is the unique minimum-burden cell for every AKS modulus, the sign split that defines it is precisely what separates the empty Williams cell from the constructible rigid cells, Popovych's variant only raises the burden, and the CM-elliptic analogue obeys the same stabilizer-index floor. A counterexample to Agrawal's conjecture, if one is ever exhibited, will require either a genuinely new subset-product solver for the low-density regime or a mechanism outside the Frobenius-matching family.

A Data tables

B	B'	j	ratio	proj. pool	demand bits
1259	12590	4	$13.8 \pm 24\%$	2.5×10^5	18,117
1259	18885	4	$41.0 \pm 26\%$	1.1×10^6	27,101
2003	20030	4	$45.5 \pm 24\%$	1.3×10^6	28,768
2003	40060	4	$209 \pm 32\%$	1.2×10^7	57,514

Table 5: Partition-first supply/demand ratios, trustworthy ($\geq$ 8-hit) $j = 4$ cells of the A2 grid, 8M samples per cell. The first row is the frozen spec. Gate G2 threshold: 2.5.

Solver frontier (odd part, synthetic instances with planted solutions): exact meet-in-the-middle $N = 24/D = 11$ instant, $N = 40/D = 15$ in 11 s, exponential beyond; annealing within the GF(2) family stalls at ≥ 19 odd components; requirement at the co-designed spec: thousands of components, $\sim$24,000 odd bits. Hierarchy census below 10^7: 105 Carmichael, 135 Lucas–Carmichael, 0 Williams, 0 order-2. Howe's rigid order-2 number: confirmed rigid at all eight primes (level-2

r	burden	minimizing structure
$5, 13, 17, 29, 37, 41$	2	$d = 4$, levels $\{1, 2\}$: the double condition
$7, 19, 31$	3	$d = 3$, levels $\{1, 3\}$: $\Phi_3(p) = p^2+p+1$ smooth
11	5	$d = 5$, levels $\{1, 5\}$
23	11	$d = 11$, levels $\{1, 11\}$

Table 6: The burden $\min_{d|r-1,\, d\geq 3}(d/2$ if even else $d)$ of Corollary 5.2 over the empirically scanned moduli; the formula reproduces the scan exactly (11/11, value and minimizing d). Popovych burdens (full depth): $4, 3, 5, 3, 4, 3, 11, 4, 3, 3, 4$ for $r = 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41$.

Frobenius powers all even), not a Williams number, $n \equiv 1 \pmod 5$, and $T(-1, n, 5)$ *fails* (script `ag3_hierarchy_check.py`, checks H1/H4).

B Reproducibility

Every quantitative claim traces to a dated section of `data/FINDINGS.md` or `doc/arith-geom-findings.md` and to a deterministic script in `code/` with committed JSON output in `data/generated/`:

claim	script	record
machinery unit tests	`verify_machinery.py`	Phase 0
0.06 invariant	`phase1a_calibration.py`, `b547_pool_analysis.py`	Phase 1a, B547
CHM closure validation	`chm_closure.c` / `.py`	B=100/113/200
yield grid, G1	`a1_yield_prototype.py`	`a1_yield_grid.json`
ratio grid + validation, G2	`a2_grid.py`, `a2_validate_and_extend.py`	A2
2-part / decomposition	`a4_solver.py`	A4
co-design window	`a5_codesign.py`, `a5_yield.py`	A5
generic-solver frontier	`a6_oddsolver.py`	A6
ω-solver scaffold	`a7_lohniebuhr.py`, `a8_omega_solver.py`	A7/A8
N1 divisor harvest	`a9_divisor_harvest.py`	N1
A10 Wagner reach law	`a10_wagner_density.py`	A10
general-r scan	`a11_general_r.py`	A11
dimension count checks	`ag2_torus_check.py`	`ag2_torus_check.json`
hierarchy / Howe / census	`ag3_hierarchy_check.py`	`ag3_hierarchy_check.json`
elliptic cells / floors	`ag4_elliptic_check.py`	`ag4_elliptic_check.json`
Popovych burden jump	`ag5_popovych_check.py`	`ag5_popovych_check.json`

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